

Notes on Higher Topos Theory and Stacks

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Abstract su topos theory e stacks.

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1 Introduction

2 Grothendieck topologies

Let us start by recalling what a Grothendieck topology on an arbitrary category $\mathcal{C}$ is. First, recall that a *sieve* on $\mathcal{C}$ is a full subcategory $S \subset \mathcal{C}$ such that, for every morphism $f : Y \rightarrow X$ in $\mathcal{C}$, if X belongs to S , then also Y belongs to S . If $X \in \mathcal{C}$, then a *sieve on X* is a sieve on the overcategory $C_{/X}$, which we usually denotes again by the letter S . More concretely, a sieve on X is a right ideal of morphisms with codomain X , that is

$$f \in S \implies f \circ g \in S,$$

for all g such that the composition makes sense.

Let $f : Y \rightarrow X$ be a morphism in the category $\mathcal{C}$. If S is a sieve on the object X , then we can define the *pullback of the sieve S* to be a sieve on Y given by

$$f^*(S) := \{g \in C_{/Y} \mid f \circ g \in S\}.$$

If $R = \{f_i : X_i \rightarrow X\}_{i \in I}$ is a collection of objects in $C_{/X}$, then there exists a smallest sieve $S \subset C_{/X}$ which contains each f_i . This is the full subcategory $S \subset C_{/X}$ given by

$$S := \{f_i g \mid \text{dom}(f_i) = \text{cod}(g), \text{ for some } i \in I\}.$$

We will say that S is *generated by the morphisms $R = \{f_i\}_{i \in I}$* , and we will write $S = (R)$. In particular, we will say that a sieve on X is *finitely generated* if it is generated by a finite collection of morphisms in $C_{/X}$.

Definition 2.1. Let C be a category. A *Grothendieck topology* τ on C is a map which assigns to each object $X \in C$ a collection of sieves $\tau(X)$ such that, for each $X \in C$, the following conditions hold:

- (a) the maximal sieve $C_{/X}$ is in $\tau(X)$;
- (b) if $S \in \tau(X)$ and $f : Y \rightarrow X$, then the pullback sieve $f^*(S) \in \tau(Y)$;
- (c) if $S \in \tau(X)$, and if R is an arbitrary sieve on X such that the pullback $f^*(R) \in \tau(Y)$, for every $f : Y \rightarrow X$ in S , then $R \in \tau(X)$.

In the following, if $S \in \tau(X)$, we will say that S is a *covering sieve* of X or, simply, that S *covers* X . A category C endowed with a Grothendieck topology will be called a *site*, and will be denoted by (C, τ) .

Example 2.2. The quintessential and motivating example of a Grothendieck topology is the one defined on topological spaces or, more precisely, on the category $\mathbf{Open}(X)$ of open subsets of $X \in \mathbf{Top}$. Indeed, a sieve S over an open subset $U \in \mathbf{Open}(X)$ is nothing but a collection of open subsets of U such that if $V' \subseteq V \in S$, then also $V' \in S$. A Grothendieck topology on $\mathbf{Open}(X)$ can be defined by requiring that a sieve covers the open subsets U if U is the union of all subsets in S .

The above example shows an important detail. In general topology, an open cover of an open subset U of X is just a family $\{U_i \mid i \in I\}$ such that $U = \cup_i U_i$, but this is not necessarily a covering sieve as defined above. However, it does generate such a sieve for each $i \in I$, that is

$$S := \{V \subset U \mid V \subseteq U_i\}.$$

Noting that the pullback in $\mathbf{Open}(X)$ is just the intersection, we can generalize this situation.

Definition 2.3. Let C be a category with all pullbacks, a *basis for a Grothendieck topology* is a map k which assigns to each object $X \in C$ a collection $k(X)$ of families of objects of $C_{/X}$ such that the following conditions hold:

- (a) if $f : Y \rightarrow X$ is an isomorphism, then the singleton $\{f\}$ is in $k(X)$;
- (b) if $\{f : Y_i \rightarrow X\}_{i \in I} \subset k(X)$, and $g : Y \rightarrow X$, then $\{Y \times_X Y_i \rightarrow Y\}_{i \in I}$ is in $k(Y)$;
- (c) if $\{f : Y_i \rightarrow X \mid i \in I\} \subset k(X)$, and for each $i \in I$ we have a family $\{g_{ij} : Z_j \rightarrow Y_i \mid j \in I_i\}$ in $k(Y_i)$, then the family $\{f_i \circ g_{ij} : Z_j \rightarrow X \mid i \in I, j \in I_i\}$ is in $k(X)$.

In the following, an element of $k(X)$ will be called a *covering family* of X . As the name suggests, if k is a basis on C , then it generates a topology τ by

$$S \in \tau(X) \iff \exists R \in k(X) \text{ such that } R \subseteq S.$$

Shortly, this means that a sieve is a covering sieve with respect to τ if and only if it contains a covering family of k (as the case of topological spaces suggested). Suppose now we have a site (C, τ) . First, note that, in general, there exist various basis for the topology τ . However, among such we can find a maximal one, that is the basis k given by

$$R \in k(X) \iff (R) \in \tau(X),$$

where (R) is the sieve generated by the family R .

Remark 2.4. In talking about Grothendieck topologies, it is customary (especially in geometry) to consider basis rather than the actual topology. Therefore, when one defines a Grothendieck topology on a category C , she/he actually defines a basis for such topology. Furthermore, one often loosely says that a family $R = \{f_i : X_i \rightarrow X\}_{i \in I}$ covers X if (R) is a covering sieve in the topology.

Mettere qualche esempio facile

3 Sheaves for a Grothendieck Topology

As we have seen in Example 2.2, Grothendieck topology represents a categorical extension of topology defined on a set. A useful construction in the latter setting is the one of sheaves on a topological space, so it is natural to ask how this notion can be extended to the case of a site. In what follows, we will consider a site $C = (C, \tau)$ and a presheaf $P : C^{op} \rightarrow \mathbf{Sets}$.

Definition 3.1. Let $X \in C$ and let $S \in \tau(X)$ be a covering sieve for X , a *matching family for P with respect to S* is an element of the limit

$$\varprojlim_{Y \in S^{op}} P(Y).$$

Remark 3.2. Some comments regarding the above definition is necessary. First of all, note that we adopted a customary abuse of notation in writing $Y \in S^{op}$, since we are actually considering the correspondent map $f \in S^{op}$ whose domain is Y . We considered the opposite of S simply to correct for the fact that a presheaf P is contravariant. Finally, writing the correspondent diagram it is easy to see that a matching family for P with respect to S is nothing but a collection of sections $\{y_f\}_{f \in S}$, each of which belongs to its correspondent $P(\text{dom}(f))$, such that given another map $g : Z \rightarrow Y$ in S , it holds that

$$P(g)(y_f) = y_{fg}.$$

For each object $X \in C$ and each covering sieve S of X , there exists a canonical map

$$\theta_S : P(X) \rightarrow \varprojlim_{Y \in S^{op}} P(Y) \tag{1}$$

which restricts a global section to its subsections. Indeed, let $x \in P(X)$ be a section and define $\theta_S(x) = \{x_f\}_{f \in S}$ by restricting x along each $f : Y \rightarrow X$, that is

$$x_f := P(f)(x) \in P(Y).$$

This family is actually a matching family for P with respect to S because, given any composable morphism $g : Z \rightarrow Y$ (so $f \circ g \in S$),

$$P(g)(x_f) = P(g)(P(f)(x)) = P(fg)(x) = x_{fg}.$$

Thus the family $\{x_f\}_{f \in S}$ is indeed an element of the limit and we can therefore define

$$\theta_S(x) := \left(P(f)(x) \right)_{f \in S}.$$

Definition 3.3. Let $X \in C$ and let $S \in \tau(X)$ be a covering sieve for X . If $m = \{x_f\}_{f \in S}$ is a matching family for P with respect to S , an *amalgamation* for m is a section $x \in P(X)$ such that $x_f = P(f)(x)$, for each $f \in S$.

The above discussion can be rephrased as follows. Given an object $X \in C$ and a covering sieve S of X , every global section $x \in P(X)$ is an amalgamation for the matching family given by its subsections (specified by the covering sieve). However, recalling the local-to-global property of a sheaf over a topological space, what we would like to conclude is the inverse implication, *i.e.* that given a matching family of P with respect to S we can always find an amalgamation, and we would like this amalgamation to be unique. All this boils down to the following definition.

Definition 3.4. A presheaf P on a site C is said to be a *sheaf* if, for each object $X \in C$ and each covering sieve $S \in \tau(X)$, the canonical map 1 is a bijection. We will denote by $\mathbf{Sh}(C)$ the full subcategory of $\mathbf{Fun}(C^{op}, \mathbf{Sets})$ consisting of sheaves over C .

Remark 3.5. The above definition of a sheaf over a site can be expressed in a form similar to the one usually given for a topological space. Recalling that in $\mathbf{Sets}$ a limit is, by definition, an equalizer (see pag. 113 of [1]) then the above discussion implies that P is a sheaf on C if, and only if, for each $X \in C$ and each covering sieve $S \in \tau(X)$, the diagram

$$P(X) \xrightarrow{e} \prod_{f \in S} P(\text{dom}(f)) \xrightleftharpoons[q]{p} \prod_{f, g: f \in S} P(\text{dom}(g))$$

is an equalizer of sets. Here $e(x) = \{P(f)(x)\}_{f \in S}$, the map f ranges over S and g are maps composable with f (hence $fg \in S$), while $p(x)_{f, g} = P(fg)(x)$ and $q(x)_{f, g} = P(g)(x_f)$.

Example 3.6. If we consider the site consisting on the category $\mathbf{Open}(X)$, for a given topological space X , together with the Grothendieck topology of Example 2.2, we obtain the category of sheaves on X , denoted as $\mathbf{Sh}(X)$.

If the category C is small, we can extend the following result, which holds for sheaves on a topological space X .

Theorem 3.7. Let C be a small category endowed with a Grothendieck topology τ . Then, there exists an adjunction

$$L : \mathbf{Fun}(C^{op}, \mathbf{Sets}) \rightleftarrows \mathbf{Sh}(C) : \iota, \quad (2)$$

where $\iota : \mathbf{Sh}(C) \rightarrow \mathbf{Fun}(C^{op}, \mathbf{Sets})$ is the inclusion, while $L : \mathbf{Fun}(C^{op}, \mathbf{Sets}) \rightarrow \mathbf{Sh}(C)$ is the *sheafification functor*. Moreover, L preserves finite limits.

Proof. See [maclane2012sheaves]

q.e.d.

Recall that if C is a locally small category, the Yoneda embedding $C \ni X \mapsto h_X(_) = \text{Hom}_C(_, X) \in \mathbf{Fun}(C^{op}, X)$ allows us to see the category C as a particular full subcategory of its category of presheaves. Given a Grothendieck topology on C , we could ask if the correspondent presheaf h_X are actually a sheaf, for each $x \in C$. This situation deserves a name.

Definition 3.8. A Grothendieck topology on a locally small category C is said to be *subcanonical* if, for every objects $X \in C$, the functor $h_X(_) = \text{Hom}_C(_, X)$ is a sheaf.

Note that if C is a locally small category endowed with a subcanonical Grothendieck topology, then the Yoneda embedding is actually a fully faithful embedding $h : C \rightarrow \mathbf{Sh}(C)$.

Remark 3.9. If C is a locally small category endowed with a non-subcanonical Grothendieck topology, by applying the functor L of Theorem 2 to the presheaf h_X , for $X \in C$, we can define the correspondent sheaf, which we denote by $\tilde{h}_X$. Thus, we can also define a *sheafified Yoneda embedding* by $X \mapsto \tilde{h}_X$. However, note that this functor is not fully faithful, it is only if the Grothendieck topology is subcanonical.

We now discuss three important examples of Grothendieck topologies and sheaves which, in the meantime, allows us to introduce notions that will be used extensively in the following section.

3.1 The Regular Topology

Definition 3.10. Let C be a category which admits pullbacks. We say that a morphism $f : X \rightarrow Y$ in C is an *effective epimorphism* if it exhibits Y as the coequalizer of the projection maps $\pi_1, \pi_2 : X \times_Y X \rightrightarrows X$.

Remark 3.11. Clearly, every effective epimorphism is also an epimorphism, while the converse need not be true. In particular, note that while $f : X \rightarrow Y$ being an epimorphism means that the induced map $\text{Hom}_C(Y, Z) \rightarrow \text{Hom}_C(X, Z)$ is injective for each $Z \in C$ (so that $\text{Hom}_C(Y, Z)$ is actually a subclass of $\text{Hom}_C(X, Z)$), being an effective epimorphism allows us to characterize this subclass, *i.e.* $\text{Hom}_C(Y, Z)$ are given by those maps $g : X \rightarrow Z$ such that the following diagram commutes

$$\begin{array}{ccccc}
 & & X & & \\
 & \nearrow & & \searrow & \\
 X \times_Y X & & & & Y \dashrightarrow Z \\
 & \searrow & & \nearrow & \\
 & & X & &
 \end{array}
 \quad
 \begin{array}{c}
 g \\
 \\
 \\
 \\
 g
 \end{array}$$

Lemma 3.12. Let C be a category which admits pullbacks, and suppose effective epimorphisms in C are stable by pullbacks. Then, effective epimorphisms in C are stable by compositions.

Proof. da scrivere

q.e.d.

Definition 3.13. A category C which admits pullbacks is said to be *regular* if

- (a) it admits finite limits;
- (b) every morphism $f : X \rightarrow Y$ can be decomposed as $f = m \circ e$, where e is an effective epimorphism and m a monomorphism;
- (c) the collection of effective epimorphisms is stable under pullbacks.

Definition 3.14. Let C be a regular category and let $X \in C$. A sieve $S \in C_{/X}$ is a *regular sieve* if it contains an effective epimorphism.

Let C be a regular category. Exploiting the above results and noting that, for each object X in C , the identity morphism id_X is an effective epimorphism, we can easily conclude the following result

Lemma 3.15. If C is a regular category, the collection of regular sieves determines a Grothendieck topology on C . We will call this topology the *regular topology* on C .

3.2 The Extensive Topology

Definition 3.16. Let C be a category which admits pullbacks, and let $X, Y \in C$ such that $X \amalg Y$ exists in C . We say that $X \amalg Y$ is a *disjoint coproduct* of X and Y if it holds that:

- (a) both the inclusion maps $\iota_X : X \rightarrow X \amalg Y$ and $\iota_Y : Y \rightarrow X \amalg Y$ are monomorphisms;
- (b) the pullback $X \times_{X \amalg Y} Y$ is an initial object of C .

Example 3.17. mettere un esempio/osservazione.

Definition 3.18. A category C which admits finite limits is said to be *extensive* if

- (a) it admits finite coproducts and coproducts in C are disjoint;
- (b) for every morphism $f : X \rightarrow Y$ in C , the pullback functor

$$f^* : C_{/Y} \rightarrow C_{/X},$$

which sends a map $\alpha : Z \rightarrow Y$ in $f^*(\alpha) : Z \times_Y X \rightarrow X$, preserves finite coproducts.

3.3 The Coherent Topology

4 Grothendieck topoi and geometric morphisms

Definition 4.1. A category $\mathcal{C}$ is said to be a *Grothendieck topos* if there exists a site (C, τ) such that $\mathcal{C}$ is equivalent to $\mathbf{Sh}(C)$.

The above definition is of limited practical use. However, a remarkable theorem of J. Giraud shows that Grothendieck topoi may be completely characterized by categorical requirements on $\mathcal{C}$. In order to state this result, we need some preliminary definitions.

5 Towards localic topoi: preliminaries

References

1. MAC LANE, S.: *Categories for the Working Mathematician*. 2nd ed. New York: Springer, 2013. Graduate Texts in Mathematics.